

Jahnavi Nalla

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EDUCATION

Arizona State University, Tempe Campus | GPA: 4.00 / 4.00

January 2024- December 2025

Master of Science in Information Technology Engineering

- Relevant Courses: Data structures, System Design, Adv. Database Management, Cloud Architecture, AI in cybersecurity, Cloud Security and Ops, Network Forensics, Security Compliance for IT.

B V Raju University, Hyderabad campus | GPA : 8.91 / 10.00

August 2019- May 2023

Bachelor of Science in Computer Science

CORE COMPETENCIES

Languages:	TypeScript, JavaScript (ES6+), C#, Java, Python, SQL
Frontend:	React 19, Angular, HTML5, CSS3
Backend & APIs:	.NET, Node.js, Hapi.js, REST APIs, GraphQL, SOAP, OAuth 2.0, JWT, RBAC, Swagger/Open API, Joi, Spring Boot
Cloud & DevOps:	AWS (Lambda, S3, DynamoDB, SQS, EC2, CDK), Azure, Docker, Kubernetes, Terraform, GitLab CI/CD, Git
Databases:	Microsoft SQL Server, PostgreSQL, MySQL, MongoDB, Supabase
AI Automation:	OpenAI SDK, LangGraph, LLM integration (Claude, GPT), Copilot, Multi-agent AI systems, Prompt optimization.
Testing & Practices:	Vitest, Jest, Postman, Agile/Scrum, code reviews, JIRA.

WORK EXPERIENCES

Arrant Technologies

Dallas, TX

Software Development Engineer

Jan 2026 - Present

- Contributed to the design and development of a cleanroom monitoring platform used by an industrial environmental-monitoring client to track environmental conditions like differential pressure, temperature, and humidity against ISO standards, with a **React 19 + TypeScript + Redux** front end and a **Node.js(Hapi.js) REST API** backend
- Built the ingestion pipeline streaming sensor readings from client facilities through AWS Lambda and SQS into DynamoDB, batching SQS consumption to raise consumer throughput and keep compliance dashboards near-real-time.
- Wrote contract-first Open API specifications with Joi schemas enforced on Hapi routes, rejecting malformed requests at the handler and giving client and server teams a single source of truth.
- Built Postman integration and regression suites that run in GitLab CI on every merge request, cutting manual pre-release API testing from 4 hours to 30 minutes.
- Wrote unit tests in Jest for shared UI components and API service handlers, covering error paths and schema edge cases.
- Integrated the OpenAI SDK to power in-product recommendations, adding prompt guardrails and response validation so model output is checked before it reaches users.

INTERSHIP EXPERIENCES

Software Development Engineer Intern | Arrant Technologies

May 2025 – August 2025

- Built reusable React + TypeScript UI components on an Agile team, incorporating feedback from code reviews with senior engineers.
- Fixed bugs in backend REST endpoints, wrote endpoint documentation, and built Postman suites validating server responses against expected schemas and status codes.
- Implemented Redux Toolkit state management for dashboard and result pages, mapping backend data models to frontend state.
- Applied WCAG accessibility guidelines - keyboard navigation, ARIA labels, color contrast to shipped UI components.

Software Engineer Intern | GVR info Systems

May 2023- December 2023

- Spearheaded the development and successful launch of a feature that optimized the submission process for store hours change requests by club associates, achieving a 40% increase in operational efficiency.
- Pioneered a streamlined mobile interface using React Native for iOS, reducing request submission time by 80%.
- Engineered RESTful APIs using Spring MVC to handle requests and approvals; optimized database interactions with Spring Data JPA enhancing data retrieval efficiency. Collaborated with UX designers to incorporate feedback

PERSONAL PROJECTS

PRISMA (Demand Forecasting & Supply-Chain Intelligence)

August 2025 –December 2025

- Built the backend in C#/ .NET- a REST API over Microsoft SQL Server handling inventory data ingestion, validation, and transactional logic for factory supply-chain records.
- Integrated time-series foundation models (Chronos, TimesFM) with ensemble learning for demand forecasting on historical factory supply-chain metrics, improving MAPE by 40% over a seasonal-naive baseline.

CHIRP - Full-Stack Social Media Platform - Personal Project

Dec 2025

- Built a Twitter-style social platform as a pnpm monorepo, used gRPC with Protocol Buffers for service-to-service communication between backend microservices, and exposed JSON REST endpoints for the browser-facing admin dashboard - gRPC for type safety and low-latency internal calls, REST for browser compatibility.
- Designed the API layer around Protocol Buffers with code-generated gRPC contracts shared across frontend and backend, ensuring end-to- end type safety, plus database migration and seeding workflows for reproducible local setup.
- Unit-tested gRPC service handlers with Vitest and integration-tested REST endpoints with Supertest in CI.